

ManipEvalAgent: Promptable and Efficient Evaluation Framework for Robotic Manipulation Policies

Yiteng Chen*, Huiping Zhuang*, Wenbo Li*, Shiyi Wang, Xiangyu Zhao, Qingyao Wu*

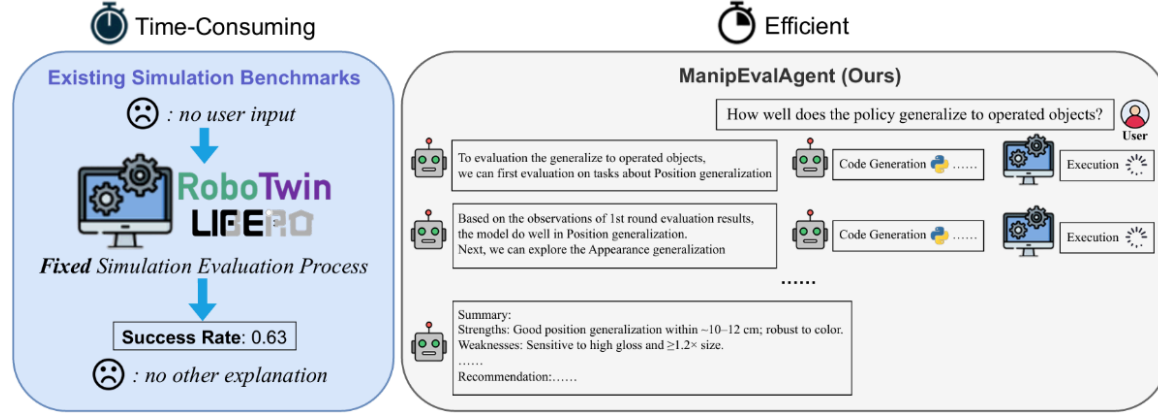


TL;DR

Agentic layer over existing simulation benchmarks, enabling promptable and interpretable evaluation.

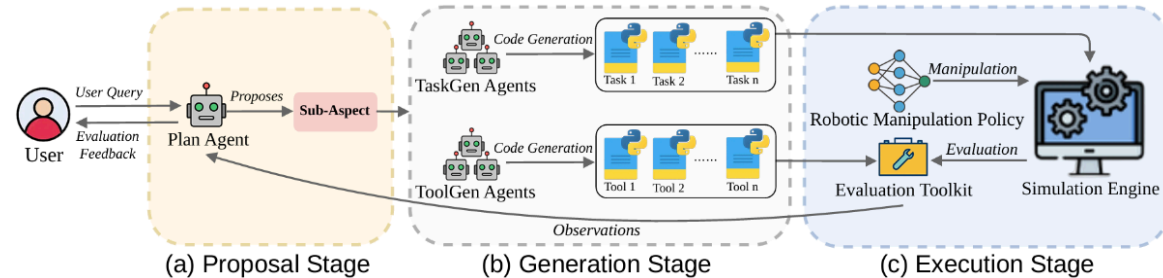
Motivation

Current simulation benchmarks usually require exhaustive rollouts, offer little customization, and provide only success rates, whereas human experts can form efficient and fine grained judgments from a few trials.



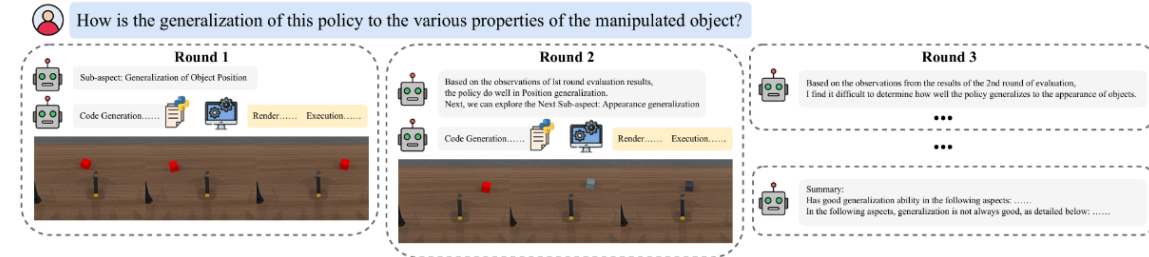
Method

ManipEvalAgent is a three-stage framework that plans evaluation aspects, generates tasks and evaluation tools in simulation, and executes multi-round assessment with adaptive refinement.



Qualitative Results

ManipEvalAgent enables multi-round, adaptive, and agentic evaluation driven by open ended user queries.



Quantitative Results

ManipEvalAgent substantially reduces evaluation time while maintaining generally comparable conclusions to existing simulation benchmarks

Table 1: Compared with existing simulation benchmarks, ManipEvalAgent significantly reduces the overall evaluation time across multiple robot manipulation policies.

Models	RoboTwin	LIBERO	Ours
ACT	167 min, 56592 samples	117 min, 29546 samples	42 min, 16927 samples
DP	171 min, 55551 samples	132 min, 29059 samples	45 min, 16895 samples
DP3	159 min, 52087 samples	113 min, 28343 samples	44 min, 15638 samples
RDT	210 min, 55435 samples	132 min, 28878 samples	63 min, 16676 samples
π_0	164 min, 51087 samples	103 min, 26732 samples	43 min, 15336 samples

Table 2: We compare the consistency of conclusions between ManipEvalAgent and existing simulation benchmarks across multiple capability dimensions. Across ten trials of the ManipEvalAgent, the percentage of results falling within the exact range (left) or within the error margin (right) is shown.

Dimension	ACT	DP	DP3	RDT	π_0
S.R. (RoboTwin)	50% / 90%	60% / 100%	50% / 80%	50% / 60%	70% / 100%
S.R. (LIBERO Avg.)	60% / 70%	50% / 70%	40% / 60%	70% / 90%	50% / 50%
Spatial (LIBERO)	70% / 100%	100% / 100%	80% / 80%	70% / 100%	60% / 80%
Obj (LIBERO)	60% / 80%	50% / 70%	60% / 60%	60% / 60%	40% / 70%
Goal (LIBERO)	30% / 70%	70% / 70%	50% / 70%	50% / 60%	50% / 50%
Long (LIBERO)	60% / 70%	60% / 80%	50% / 70%	70% / 80%	60% / 90%

Table 9: Policy ranking consistency between standard Simulation benchmarks and ManipEvalAgent. We report Spearman's rank correlation ρ with bootstrap confidence intervals (90% CI for 10 rollouts; 95% CI for 20 and 50 rollouts).

Setting	10 rollouts ρ (90% CI)	20 rollouts ρ (95% CI)	50 rollouts ρ (95% CI)
RoboTwin (short, 0-500)	0.83 [0.75, 0.90]	0.79 [0.65, 0.90]	0.81 [0.70, 0.90]
RoboTwin (medium, 600-1000)	0.86 [0.70, 0.90]	0.81 [0.60, 0.90]	0.83 [0.75, 0.90]
RoboTwin (long, 1100+)	0.83 [0.65, 0.90]	0.77 [0.70, 0.90]	0.80 [0.65, 1.00]
LIBERO-Object	0.80 [0.70, 1.00]	0.91 [0.70, 1.00]	0.82 [0.70, 0.90]
LIBERO-Spatial	0.76 [0.65, 0.85]	0.81 [0.65, 0.90]	0.82 [0.60, 0.90]
LIBERO-Goal	0.81 [0.70, 0.90]	0.73 [0.60, 0.85]	0.79 [0.70, 0.90]
LIBERO-Long	0.85 [0.70, 0.90]	0.87 [0.75, 1.00]	0.85 [0.80, 0.90]

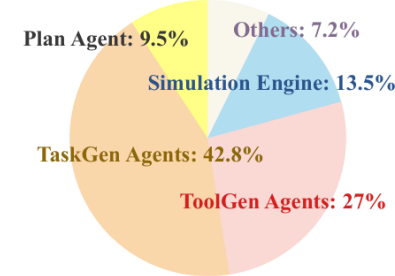


Figure 6: System Error Breakdown

Future Directions

Expand ManipEvalAgent from generating tasks to also generating manipulable assets within simulation. Scale ManipEvalAgent to end to end automated robotic manipulation policy development. Move ManipEvalAgent beyond simulation toward real world deployment.